CHAPTER 10 · DERIVATIONS

KINETIC THEORY OF GASES

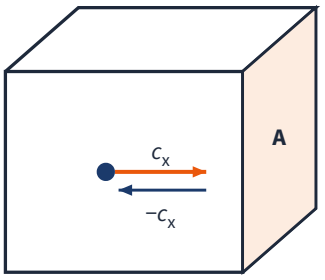
3 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 PRESSURE OF AN IDEAL GAS

SYLLABUS 10.2(C)

RESULT $p = \frac{1}{3}\rho\langle c^2\rangle$

1. N molecules, each of mass m, in a cube of side L. One molecule moves with velocity component c_x towards face A.
2. It rebounds elastically: change in momentum = $-2mc_x$, so it gives $2mc_x$ to the wall.
3. Time between hits on face A = $\frac{2L}{c_x}$.
4. Force = rate of change of momentum = $2mc_x / \frac{2L}{c_x} = \frac{mc_x^2}{L}$.
5. All N molecules: $F = (Nm/L)\langle c_x^2\rangle$; pressure $p = \frac{F}{L^2} = (Nm/V)\langle c_x^2\rangle$.
6. Random motion: $\langle c^2\rangle = \langle c_x^2\rangle + \langle c_y^2\rangle + \langle c_z^2\rangle = 3\langle c_x^2\rangle$.
7. **$p = \frac{1}{3}(Nm/V)\langle c^2\rangle = \frac{1}{3}\rho\langle c^2\rangle$**, where $\rho = Nm/V$.



face A
$\Delta p = 2mc_x$

2 MEAN TRANSLATIONAL KINETIC ENERGY

SYLLABUS 10.3(C)

RESULT $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$

1. From the kinetic theory: $pV = \frac{1}{3}Nm\langle c^2\rangle$.
2. Ideal gas: $pV = nRT = NkT$ ($k = \frac{R}{N_A}$).
3. $\frac{1}{3}Nm\langle c^2\rangle = NkT$, so $m\langle c^2\rangle = 3kT$.
4. **$\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$**: the mean KE depends only on the temperature.

3 INTERNAL ENERGY AND DEGREES OF FREEDOM

SYLLABUS 10.5(B)

RESULT $U = \frac{f}{2}nRT$

1. Ideal gas: no intermolecular forces, so the internal energy is the total kinetic energy of the molecules.
2. Equipartition: each degree of freedom has mean energy $\frac{1}{2}kT$ per molecule; a molecule with f degrees of freedom has $\frac{1}{2}fkT$.
3. N molecules: $U = \frac{1}{2}fNkT = $ **$\frac{f}{2}nRT$**.
4. Monatomic f = 3: $U = \frac{3}{2}nRT$. Diatomic at room temperature f = 5: $U = \frac{5}{2}nRT$.